

The results below are generated from an R script.

```
#code to calculate the alpha diversity
setwd("~/Desktop/Screenshots/Career/Vollmer Lab/GitHub/16S_Florida_Tank_Analysis/Code")

#### Packages ####
library(phyloseq)
library(microbiome)

## Loading required package: ggplot2
## Need help getting started? Try the R Graphics Cookbook: https://r-graphics.org
##
## microbiome R package (microbiome.github.com)
##
##
## Copyright (C) 2011-2022 Leo Lahti,
## Sudarshan Shetty et al. <microbiome.github.io>
##
## Attaching package: 'microbiome'
## The following object is masked from 'package:ggplot2':
##
## alpha
## The following object is masked from 'package:base':
##
## transform

library(vegan)

## Loading required package: permute
## Loading required package: lattice
## This is vegan 2.6-4
##
## Attaching package: 'vegan'
## The following object is masked from 'package:microbiome':
##
## diversity

library(lme4)

## Loading required package: Matrix

library(afex)

## *****
## Welcome to afex. For support visit: http://afex.singmann.science/
## - Functions for ANOVAs: aov_car(), aov_ez(), and aov_4()
## - Methods for calculating p-values with mixed(): 'S', 'KR', 'LRT', and 'PB'
## - 'afex_aov' and 'mixed' objects can be passed to emmeans() for follow-up tests
## - NEWS: emmeans() for ANOVA models now uses model = 'multivariate' as default.
## - Get and set global package options with: afex_options()
## - Set orthogonal sum-to-zero contrasts globally: set_sum_contrasts()
## - For example analyses see: browseVignettes("afex")
## *****
##
## Attaching package: 'afex'
```

```

## The following object is masked from 'package:lme4':
##
##      lmer

library(emmeans)
library(car)

## Loading required package: carData

library(emmeans)
library(multcomp)

## Loading required package: mvtnorm
## Loading required package: survival
## Loading required package: TH.data
## Loading required package: MASS
##
## Attaching package: 'TH.data'
## The following object is masked from 'package:MASS':
##
##      geyser

library(tidyverse)

## - Attaching packages ----- tidyverse 1.3.2 -
## v tibble 3.1.8      v dplyr 1.1.0
## v tidyr 1.3.0      v stringr 1.5.0
## v readr 2.1.3      v forcats 1.0.0
## v purrr 1.0.1
## - Conflicts ----- tidyverse_conflicts() -
## x microbiome::alpha() masks ggplot2::alpha()
## x tidyr::expand()      masks Matrix::expand()
## x dplyr::filter()      masks stats::filter()
## x dplyr::lag()          masks stats::lag()
## x tidyr::pack()         masks Matrix::pack()
## x dplyr::recode()       masks car::recode()
## x dplyr::select()       masks MASS::select()
## x purrr::some()         masks car::some()
## x tidyr::unpack()       masks Matrix::unpack()

#### Read in Data ####
aggregation_level <- 'Genus' #or none

microbiome_data <- read_rds("../intermediate_files/preprocess_microbiome.rds")
metadata <- sample_data(microbiome_data) %>%
  as_tibble(rownames = 'sample_id') %>%
  select(-retain_sample)

if(aggregation_level != 'none'){
  microbiome_data <- aggregate_taxa(microbiome_data, aggregation_level)
  taxa_names(microbiome_data) <- str_replace_all(taxa_names(microbiome_data), ' |-', '_')
} else {
  taxa_names(microbiome_data) <- str_c('ASV', 1:length(taxa_names(microbiome_data)), sep = '_')
}

#data filtering step?

```

```

#must use untrimmed data set for alpha diversity measures of richness to get meaningful results
#functions depend heavily on singletons

#there are 2 samples with 5 or less species observed, might be skewing the data

#### Alpha Diversity ####
alpha_table <- microbiome::alpha(microbiome_data, index = "all") %>%
  as_tibble(rownames = 'sample_id') %>%
  inner_join(metadata, by = 'sample_id') %>%
  mutate(fragment_id = str_c(str_replace_na(exposure, 'NA'), tank, genotype, sep = '_'))

## Observed richness
## Other forms of richness
## Diversity
## Evenness
## Dominance
## Rarity

alpha_table

## # A tibble: 226 x 30
##   sample_id obser~1 chao1 diver~2 diver~3 diver~4 diver~5 diver~6 evenn~7 evenn~8 evenn~9
##   <chr>          <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 Acerv_NA~      19  22.5   1.46   0.315   0.892   4.77     1     1     0.303  0.0769
## 2 Acerv_NA~      47  57.5   7.33   0.864   2.40    9.25     3   0.999  0.623  0.156
## 3 Acerv_NA~      47  60.6   10.8   0.907   3.07   17.4     5   0.996  0.797  0.229
## 4 Acerv_NA~      24  29.6   2.35   0.575   1.21    3.78     1     1     0.379  0.0980
## 5 Acerv_NA~      19  26.5   1.46   0.315   0.828   4.19     1   0.998  0.281  0.0768
## 6 Acerv_NA~      24  31     6.41   0.844   2.42    8.04     3     1     0.761  0.267
## 7 TO_D_HOM~     142 147.    8.62   0.884   2.97   20.5     3   0.900  0.599  0.0607
## 8 TO_D_HOM~     188 213.    7.93   0.874   3.33   34.9     4   0.761  0.636  0.0422
## 9 TO_D_HOM~     176 191     9.22   0.891   3.40   31.9     4   0.798  0.657  0.0524
## 10 TO_D_HOM~    156 194.    6.96   0.856   3.36   33.8     6   0.806  0.665  0.0446
## # ... with 216 more rows, 19 more variables: evenness_evar <dbl>, evenness_bulla <dbl>,
## # dominance_dbp <dbl>, dominance_dmn <dbl>, dominance_absolute <dbl>,
## # dominance_relative <dbl>, dominance_simpson <dbl>, dominance_core_abundance <dbl>,
## # dominance_gini <dbl>, rarity_log_modulo_skewness <dbl>, rarity_low_abundance <dbl>,
## # rarity_rare_abundance <dbl>, time <chr>, exposure <chr>, tank <chr>, genotype <chr>,
## # disease_state <chr>, final_disease_state <chr>, fragment_id <chr>, and abbreviated
## # variable names 1: observed, 2: diversity_inverse_simpson, ...

#only looking at time points 3 and 7
timepoint_data <- alpha_table %>%
  filter(time %in% c('T3', 'T7'))

#### Various Models ####
lm(observed ~ (exposure + final_disease_state) * time, data = timepoint_data) %>% car::Anova(type = 2)

## Anova Table (Type II tests)
##
## Response: observed
##
##           Sum Sq Df F value    Pr(>F)
## exposure      35  1  0.0639  0.800748
## final_disease_state 42  1  0.0765  0.782454

```

```

## time                73281    1 133.4975 < 2.2e-16 ***
## exposure:time        6      1   0.0111  0.916265
## final_disease_state:time 4277    1   7.7917  0.005797 **
## Residuals           102102 186
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

glm(observed ~ (exposure + final_disease_state) * time, data = timepoint_data, family = 'poisson')

##
## Call:  glm(formula = observed ~ (exposure + final_disease_state) * time,
##          family = "poisson", data = timepoint_data)
##
## Coefficients:
##              (Intercept)                exposureH            final_disease_stateH
##                4.28385                -0.02967                -0.19742
##                timeT7            exposureH:timeT7  final_disease_stateH:timeT7
##                0.25953                0.02269                0.35262
##
## Degrees of Freedom: 191 Total (i.e. Null);  186 Residual
## Null Deviance:      2438
## Residual Deviance: 1426  AIC: 2621

linear_model <- lmer(observed ~ (exposure + final_disease_state) * time + (1 | fragment_id) +
                    (1 | tank), data = timepoint_data)
count_model_1 <- glmer(observed ~ (exposure + final_disease_state) * time +
                    (1 | fragment_id) + (1 | tank), data = timepoint_data, family = 'poisson')
car::Anova(count_model_1, type = 2) #poisson is for count data

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: observed
##
##              Chisq Df Pr(>Chisq)
## exposure          0.0796  1    0.7779
## final_disease_state 0.1467  1    0.7018
## time             844.0122  1 < 2.2e-16 ***
## exposure:time      0.3480  1    0.5553
## final_disease_state:time 51.0773  1  8.88e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(count_model_1)

## Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]
##
## Family: poisson ( log )
## Formula: observed ~ (exposure + final_disease_state) * time + (1 | fragment_id) +
##          (1 | tank)
## Data: timepoint_data
##
##      AIC      BIC    logLik deviance df.resid
##    2176    2202    -1080    2160     184
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max

```

```

## -6.0307 -1.1039 -0.0845 1.0291 6.8471
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## fragment_id (Intercept) 0.0431676 0.20777
## tank        (Intercept) 0.0007822 0.02797
## Number of obs: 192, groups: fragment_id, 96; tank, 7
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      4.26315    0.04222 100.980 < 2e-16 ***
## exposureH        -0.03813    0.07748  -0.492  0.6226
## final_disease_stateH -0.18943    0.07524  -2.518  0.0118 *
## timeT7           0.25910    0.02515  10.302 < 2e-16 ***
## exposureH:timeT7    0.02845    0.04823   0.590  0.5553
## final_disease_stateH:timeT7 0.34851    0.04876   7.147 8.88e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) expsrH fnl__H timeT7 exH:T7
## exposureH    -0.176
## fnl_dss_stH  -0.398 -0.706
## timeT7        -0.337  0.041  0.155
## expsrH:tmT7   0.039 -0.395  0.303 -0.106
## fnl_ds_H:T7  0.143  0.291 -0.401 -0.432 -0.737

count_model_nb <- glmer.nb(observed ~ (exposure + final_disease_state) * time +
                           (1 | fragment_id) + (1 | tank),
                           data = timepoint_data)
summary(count_model_nb)

## Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]
## ]
## Family: Negative Binomial(11.2013) ( log )
## Formula: observed ~ (exposure + final_disease_state) * time + (1 | fragment_id) +
## (1 | tank)
## Data: timepoint_data
##
##      AIC      BIC    logLik deviance df.resid
## 1822.2  1851.6   -902.1  1804.2     183
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -2.9411 -0.6518 -0.0587  0.5042  4.4895
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## fragment_id (Intercept) 0.0060197 0.07759
## tank        (Intercept) 0.0006657 0.02580
## Number of obs: 192, groups: fragment_id, 96; tank, 7
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)

```

```

## (Intercept)                4.27906    0.05633  75.960 < 2e-16 ***
## exposureH                  -0.03295    0.10398  -0.317 0.751310
## final_disease_stateH       -0.19929    0.10344  -1.927 0.054030 .
## timeT7                     0.26164    0.07275   3.596 0.000323 ***
## exposureH:timeT7           0.02946    0.13648   0.216 0.829094
## final_disease_stateH:timeT7 0.35295    0.13884   2.542 0.011020 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) expsrH fnl__H timeT7 exH:T7
## exposureH   -0.132
## fnl_dss_stH -0.412 -0.725
## timeT7       -0.663  0.066  0.301
## expsrH:tmT7  0.066 -0.692  0.523 -0.103
## fnl_ds_H:T7  0.286  0.512 -0.704 -0.439 -0.735

car::Anova(count_model_nb, type = 2)

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: observed
##
##              Chisq Df Pr(>Chisq)
## exposure          0.0539  1    0.81648
## final_disease_state 0.0371  1    0.84719
## time             107.1688  1    < 2e-16 ***
## exposure:time       0.0466  1    0.82909
## final_disease_state:time 6.4621  1    0.01102 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

AIC(count_model_nb, count_model_1, linear_model) #better model has smaller AIC value

##              df      AIC
## count_model_nb  9 1822.233
## count_model_1   8 2175.974
## linear_model    9 1734.934

mixed(observed ~ (exposure + final_disease_state) * time +
      (1 | fragment_id) + (1 | tank), data = timepoint_data)

## Contrasts set to contr.sum for the following variables: exposure, final_disease_state,
time, fragment_id, tank

## Fitting one lmer() model. [DONE]
## Calculating p-values. [DONE]
## Mixed Model Anova Table (Type 3 tests, S-method)
##
## Model: observed ~ (exposure + final_disease_state) * time + (1 | fragment_id) +
## Model:      (1 | tank)
## Data: timepoint_data
##              Effect      df      F p.value
## 1              exposure 1, 10.21      0.04   .847
## 2    final_disease_state 1, 91.41      0.05   .820
## 3              time 1, 93.00 131.49 *** <.001

```

```
## 4          exposure:time 1, 93.00      0.01    .907
## 5 final_disease_state:time 1, 93.00    9.57 **    .003
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#does essentially the same thing as repeated measures but can do multiple random effects
aov_4(observed ~ exposure + final_disease_state + time + (1 + time | fragment_id), data = timepoint_data)

## Converting to factor: exposure, final_disease_state
## Contrasts set to contr.sum for the following variables: exposure, final_disease_state

## Anova Table (Type 3 tests)
##
## Response: observed
##
```

	Effect	df	MSE	F	ges	p.value
## 1	exposure	1, 93	650.91	0.05	<.001	.817
## 2	final_disease_state	1, 93	650.91	0.06	<.001	.800
## 3	time	1, 93	446.96	131.49 ***	.365	<.001
## 4	exposure:time	1, 93	446.96	0.01	<.001	.907
## 5	final_disease_state:time	1, 93	446.96	9.57 **	.040	.003

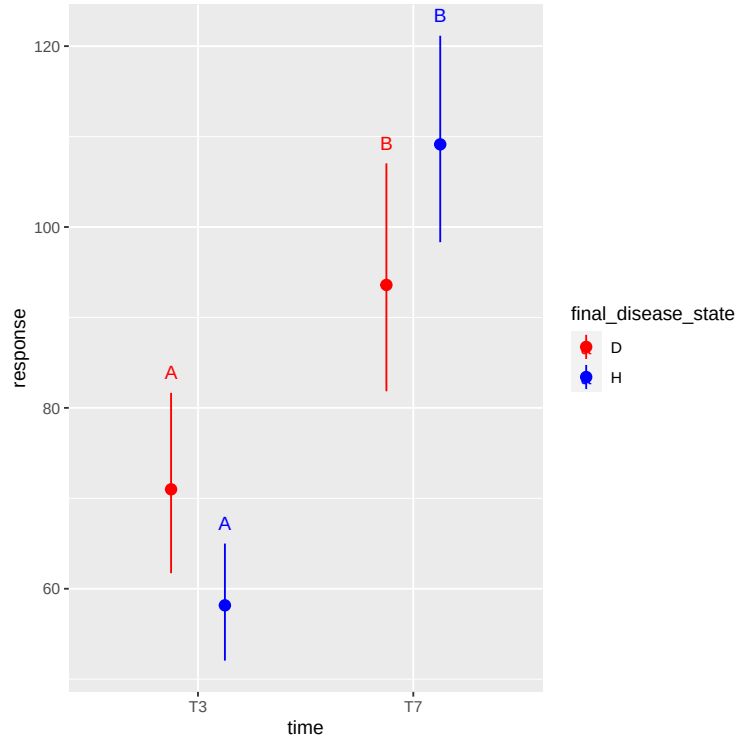
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#only one random effect at a time

emmeans(count_model_nb, ~final_disease_state * time, type = 'link') %>% contrast('pairwise')

## contrast      estimate      SE df z.ratio p.value
## D T3 - H T3      0.199 0.1034 Inf   1.927  0.2168
## D T3 - D T7     -0.276 0.0945 Inf  -2.925  0.0181
## D T3 - H T7     -0.430 0.0883 Inf  -4.870 <.0001
## H T3 - D T7     -0.476 0.0878 Inf  -5.420 <.0001
## H T3 - H T7     -0.629 0.0735 Inf  -8.566 <.0001
## D T7 - H T7     -0.154 0.0988 Inf  -1.556  0.4042
##
## Results are averaged over the levels of: exposure
## Results are given on the log (not the response) scale.
## P value adjustment: tukey method for comparing a family of 4 estimates

emmeans(count_model_nb, ~final_disease_state * time, type = 'response') %>%
  multcomp::cld(Letters = LETTERS) %>%
  as_tibble %>%
  mutate(.group = str_trim(.group)) %>%
  # rename(response = emmean,
  #       asymp.UCL = upper.CL,
  #       asymp.LCL = lower.CL) %>%
  ggplot(aes(x = time, y = response, ymin = asymp.LCL, ymax = asymp.UCL, colour = final_disease_state))
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = asymp.UCL, label = .group), position = position_dodge(0.5),
            vjust = -1) +
  scale_colour_manual(values = c('D' = 'red', 'H' = 'blue'))
```



Analyzing Alpha Table

#all diversity metrics in table

```
select(alpha_table, sample_id, observed, any_of(colnames(metadata)))
```

A tibble: 226 x 8

##	sample_id	observed	time	exposure	tank	genotype	disease_state	final_disease_state
##	<chr>	<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
##	1 Acerv_NA_D_3	19	Acerv	<NA>	D	3	<NA>	<NA>
##	2 Acerv_NA_D_4	47	Acerv	<NA>	D	4	<NA>	<NA>
##	3 Acerv_NA_D_5	47	Acerv	<NA>	D	5	<NA>	<NA>
##	4 Acerv_NA_H_3	24	Acerv	<NA>	H	3	<NA>	<NA>
##	5 Acerv_NA_H_4	19	Acerv	<NA>	H	4	<NA>	<NA>
##	6 Acerv_NA_H_5	24	Acerv	<NA>	H	5	<NA>	<NA>
##	7 T0_D_HOMO_1	142	T0	D	HOMO	1	<NA>	<NA>
##	8 T0_D_HOMO_2	188	T0	D	HOMO	2	<NA>	<NA>
##	9 T0_D_HOMO_3	176	T0	D	HOMO	3	<NA>	<NA>
##	10 T0_D_HOMO_4	156	T0	D	HOMO	4	<NA>	<NA>

... with 216 more rows

```
colnames(timepoint_data)
```

##	[1] "sample_id"	"observed"
##	[3] "chao1"	"diversity_inverse_simpson"
##	[5] "diversity_gini_simpson"	"diversity_shannon"
##	[7] "diversity_fisher"	"diversity_coverage"
##	[9] "evenness_camargo"	"evenness_pielou"
##	[11] "evenness_simpson"	"evenness_evar"
##	[13] "evenness_bulla"	"dominance_dbp"
##	[15] "dominance_dmn"	"dominance_absolute"

```

## [17] "dominance_relative"      "dominance_simpson"
## [19] "dominance_core_abundance" "dominance_gini"
## [21] "rarity_log_modulo_skewness" "rarity_low_abundance"
## [23] "rarity_rare_abundance"    "time"
## [25] "exposure"                "tank"
## [27] "genotype"                "disease_state"
## [29] "final_disease_state"     "fragment_id"

tmp <- timepoint_data %>%
  pivot_longer(cols = observed:rarity_rare_abundance,
               names_to = 'metric',
               values_to = 'value') %>%
  nest_by(metric) %>%
  summarise(model = list(lmer(value ~ (exposure + final_disease_state) * time +
                              (1 | fragment_id) + (1 | tank), data = data)))

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## 'summarise()' has grouped output by 'metric'. You can override using the '.groups' argument.

#Selected measures:

#richness - observed; dominance - dbp, gini; rarity - rare abundance
#diversity - shannon; evenness - camargo, bulla

kept_columns <- c("sample_id", "fragment_id", "observed", "dominance_dbp",
                 "dominance_gini", "rarity_rare_abundance",
                 "diversity_shannon", "evenness_camargo", "evenness_bulla")
alpha_kept_columns <- paste("alpha_table$", kept_columns, sep = "")

reduced_alpha_table <- timepoint_data %>% #only T3 and T7
  select(all_of(kept_columns), any_of(colnames(metadata)))

r_alpha_models <- reduced_alpha_table %>%
  pivot_longer(cols = observed:evenness_bulla,
               names_to = 'metric',
               values_to = 'value') %>%
  nest_by(metric) %>%
  summarise(model = list(lmer(value ~ (exposure + final_disease_state) * time +
                              (1 | fragment_id) + (1 | tank), data = data)))

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## 'summarise()' has grouped output by 'metric'. You can override using the '.groups' argument.

```

```
#### Shannon Diversity ####

#how diverse the species in a given community are

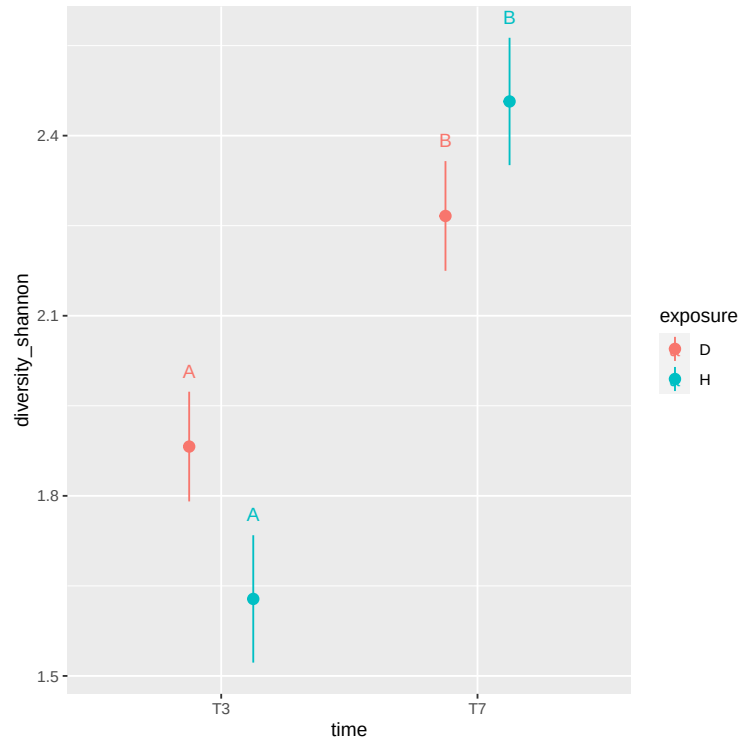
#greater diversity at T7 than T3

anova(r_alpha_models$model[[1]])

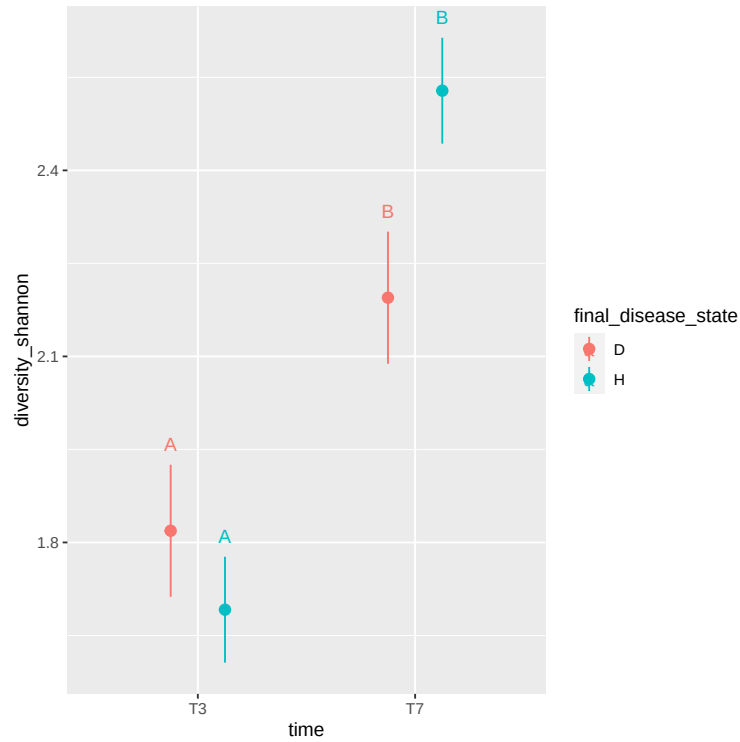
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## exposure      0.0142  0.0142     1    9.667  0.0662  0.80236
## final_disease_state 0.1771  0.1771     1   90.679  0.8244  0.36631
## time          16.3254 16.3254     1   93.000 76.0032 1.052e-13 ***
## exposure:time      1.0646  1.0646     1   93.000   4.9562  0.02841 *
## final_disease_state:time 1.1047  1.1047     1   93.000   5.1427  0.02566 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#time ***

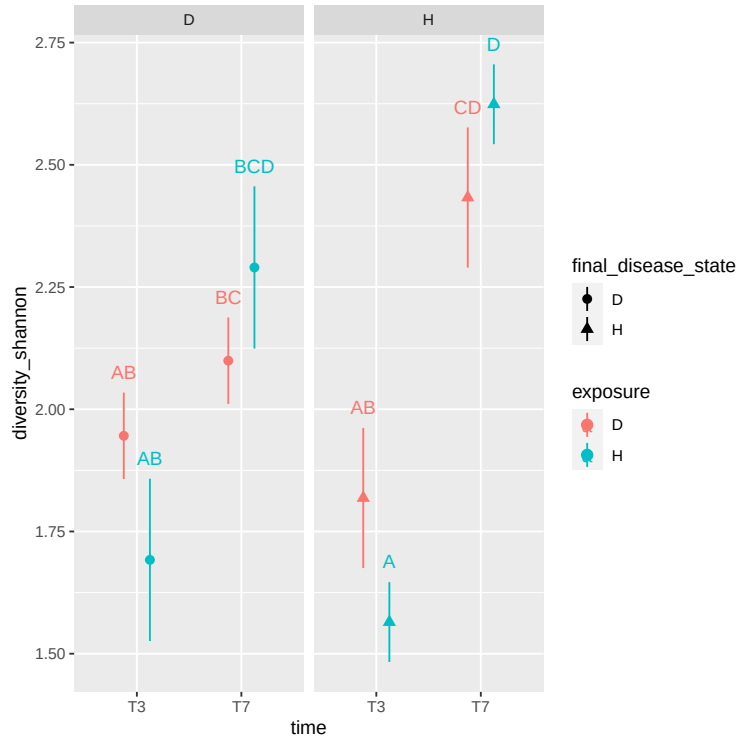
#exposure:time *
emmeans(r_alpha_models$model[[1]], ~(exposure) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
            col = exposure)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[1])
```



```
#final_disease_state:time *
emmeans(r_alpha_models$model[[1]], ~(final_disease_state) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
             col = final_disease_state)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[1])
```



```
#all
emmeans(r_alpha_models$model[[1]], ~(exposure + final_disease_state) * time,
         type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
             col = exposure, shape = final_disease_state)) +
  facet_wrap(~final_disease_state) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[1])
```



```
#### DBP Dominance ####
```

```
#relative abundance of most abundant species, 0-1 & bigger #s means more dominant
```

```
#greater dominance at T3 than T7, matches the increase in shannon diversity at T7
```

```
anova(r_alpha_models$model[[2]])
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
```

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
## exposure	0.01037	0.01037	1	186	0.4862	0.48651
## final_disease_state	0.00935	0.00935	1	186	0.4383	0.50875
## time	1.53844	1.53844	1	186	72.1426	6.301e-15 ***
## exposure:time	0.10153	0.10153	1	186	4.7612	0.03036 *
## final_disease_state:time	0.06242	0.06242	1	186	2.9270	0.08878 .

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#time ***
```

```
#final_disease_state:time .
```

```
#exposure:time *
```

```
emmeans(r_alpha_models$model[[2]], ~(exposure) * time,
```

```
  type = 'response') %>%
```

```
  cld(Letters = LETTERS) %>%
```

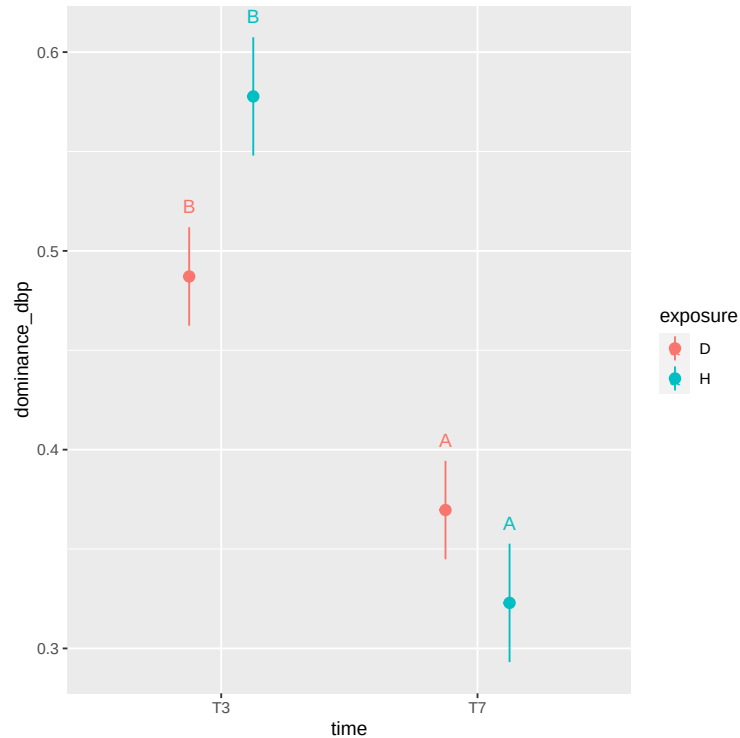
```
  as_tibble() %>%
```

```
  mutate(.group = str_trim(.group)) %>%
```

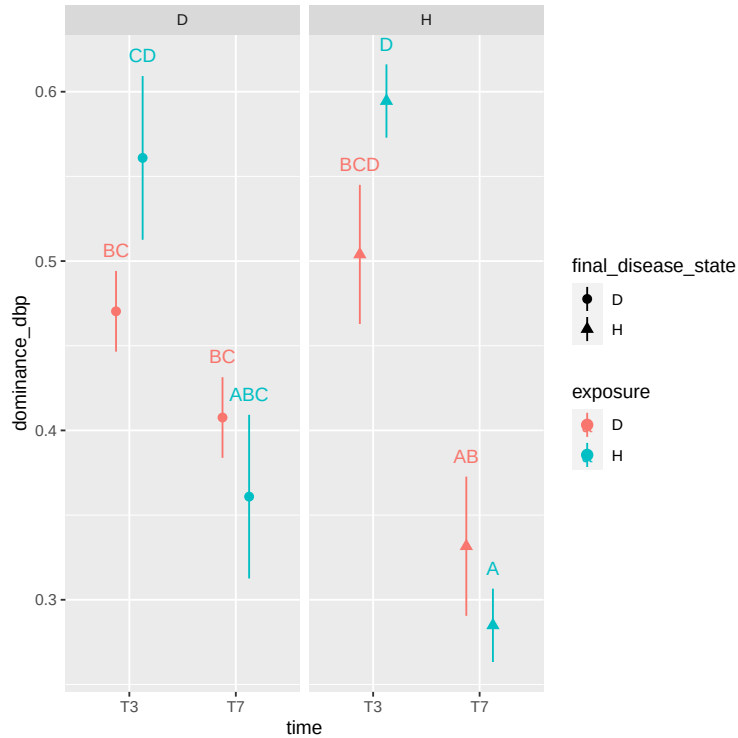
```
  #rename(emmean = response) %>%
```

```
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
             col = exposure)) +
```

```
geom_pointrange(position = position_dodge(0.5)) +
geom_text(aes(y = (emmean + SE), label = .group),
          position = position_dodge(0.5), vjust = -1) +
ylab(r_alpha_models$metric[2])
```



```
#all
emmeans(r_alpha_models$model[[2]], ~(exposure + final_disease_state) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
            col = exposure, shape = final_disease_state)) +
  facet_wrap(~final_disease_state) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[2])
```



```
#### Gini Dominance ####
```

```
#how unevenly abundances are distributed, 0-1 & perfect equality is 0
```

```
#slight increase in evenness at T7, matches the decreased dominance
#scale of y-axis is very slight
```

```
anova(r_alpha_models$model[[3]])
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
```

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
## exposure	0.0000103	0.0000103	1	3.32	0.1851	0.69342
## final_disease_state	0.0001330	0.0001330	1	90.93	2.3884	0.12571
## time	0.0037063	0.0037063	1	93.00	66.5681	1.576e-12 ***
## exposure:time	0.0001764	0.0001764	1	93.00	3.1691	0.07831 .
## final_disease_state:time	0.0003569	0.0003569	1	93.00	6.4108	0.01302 *

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#time ***
```

```
#exposure:time .
```

```
#final_disease_state:time *
```

```
emmeans(r_alpha_models$model[[3]], ~(final_disease_state) * time,
```

```
  type = 'response') %>%
```

```
  cld(Letters = LETTERS) %>%
```

```
  as_tibble() %>%
```

```
  mutate(.group = str_trim(.group)) %>%
```

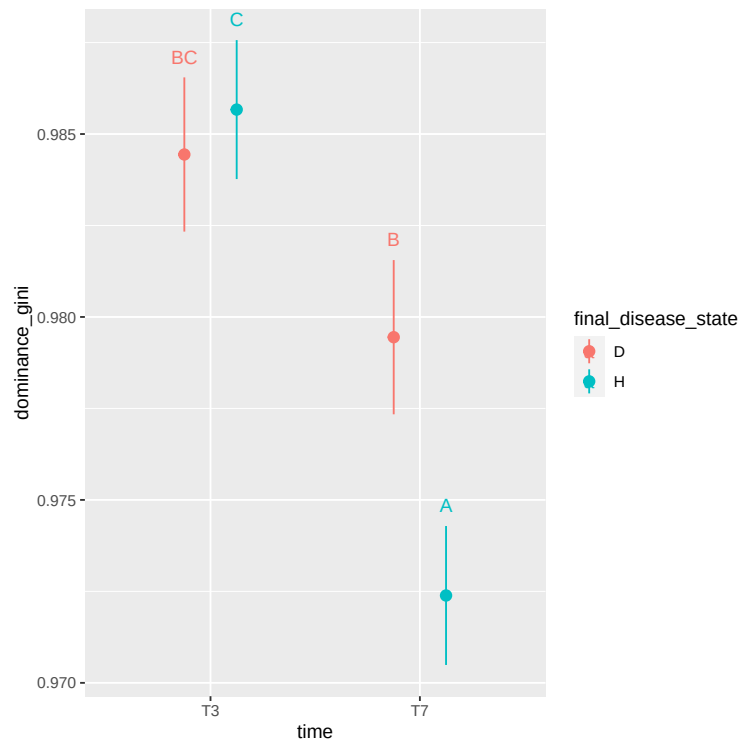
```
  #rename(emmean = response) %>%
```

```
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
```

```

    col = final_disease_state)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[3])

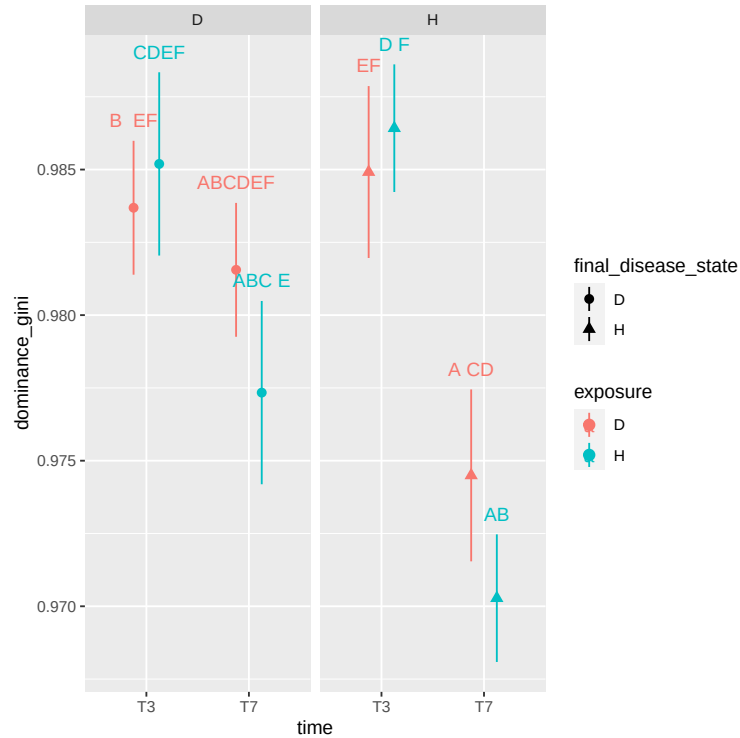
```



```

#all
emmeans(r_alpha_models$model[[3]], ~(exposure + final_disease_state) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
            col = exposure, shape = final_disease_state)) +
  facet_wrap(~final_disease_state) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[3])

```



```
#### Bulla Evenness ####
```

```
#evenness w/ equal weight to all species, sensitive to rare species
```

```
#nothing significant
```

```
anova(r_alpha_models$model[[4]])
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
```

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
## exposure	0.0004665	0.0004665	1	10.214	0.0760	0.78830
## final_disease_state	0.0011835	0.0011835	1	185.821	0.1928	0.66112
## time	0.0024538	0.0024538	1	181.511	0.3997	0.52802
## exposure:time	0.0196227	0.0196227	1	181.511	3.1966	0.07546 .
## final_disease_state:time	0.0003939	0.0003939	1	181.511	0.0642	0.80031

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#exposure:time .
```

```
    #not worth plotting (plot showed nothing)
```

```
#### Camargo's Evenness ####
```

```
#proportions of indivs between sites, 0-1 & 1 is even, 0 is patchy
```

```
#T3 is really even and T7 is patchier
```

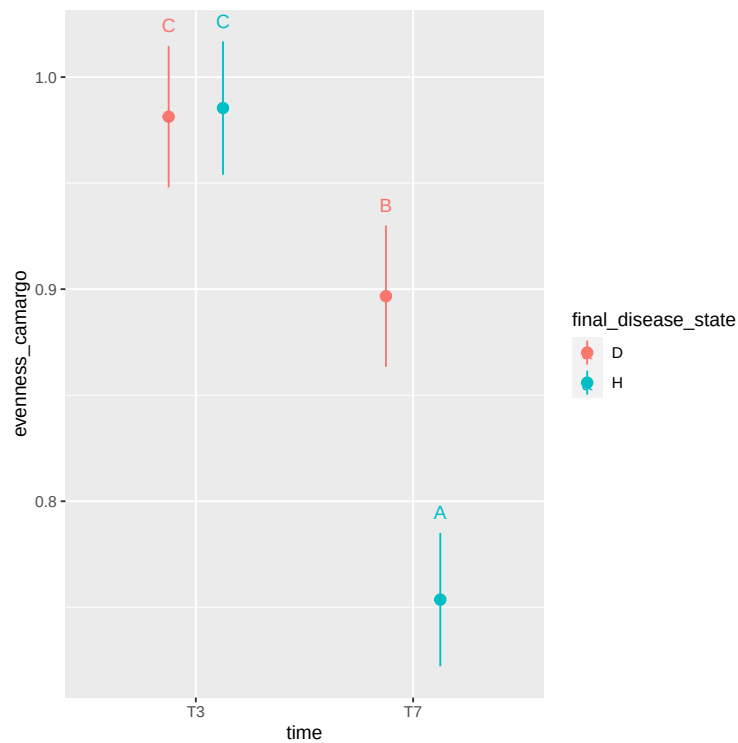
```
anova(r_alpha_models$model[[5]])
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
```

```
##               Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
## exposure      0.00012  0.00012     1    4.059   0.0107   0.922492
## final_disease_state 0.09523  0.09523     1  185.946   8.4859   0.004018 **
## time          1.11114  1.11114     1  179.628  99.0138 < 2.2e-16 ***
## exposure:time    0.01544  0.01544     1  179.628   1.3763   0.242292
## final_disease_state:time 0.11256  0.11256     1  179.628  10.0304   0.001810 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#final_disease_state **
#time ***

#final_disease_state:time **
emmeans(r_alpha_models$model[[5]], ~(final_disease_state) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
            col = final_disease_state)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[5])
```



```
#### Observed Species Richness ####
```

```
#number of species observed
```

```

#more species richness at T7 than T3, matches shannon diversity

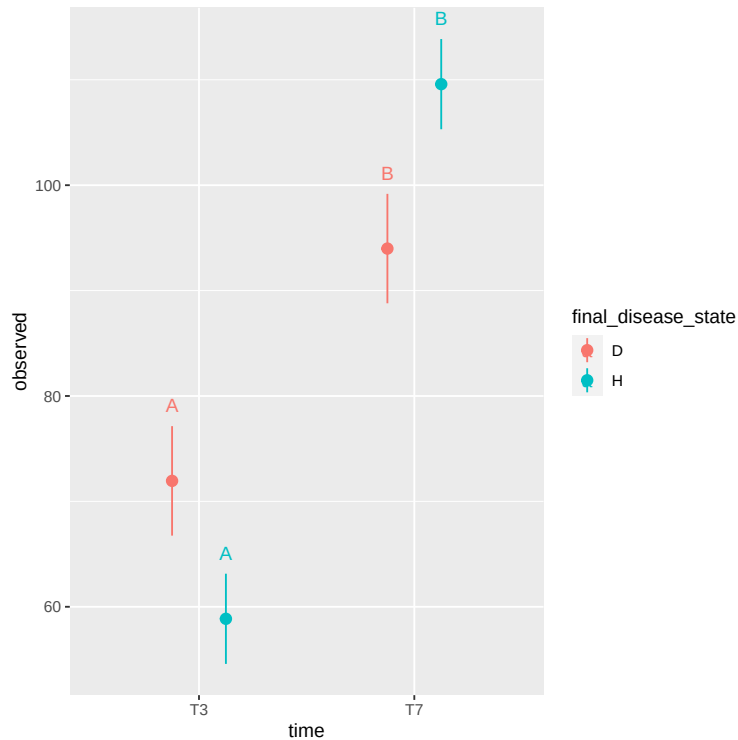
anova(r_alpha_models$model[[6]])

## Type III Analysis of Variance Table with Satterthwaite's method
##
##          Sum Sq Mean Sq NumDF   DenDF  F value    Pr(>F)
## exposure          17      17      1  10.209    0.0390  0.847327
## final_disease_state 23      23      1  91.407    0.0521  0.820010
## time          58771  58771      1  93.000  131.4910 < 2.2e-16 ***
## exposure:time         6       6      1  93.000    0.0136  0.907368
## final_disease_state:time 4277   4277      1  93.000    9.5694  0.002613 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#time ***

#final_disease_state:time **
emmeans(r_alpha_models$model[[6]], ~(final_disease_state) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
            col = final_disease_state)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[6])

```



```
#### Rare Abundance ####

#relative proportion of rare species(i.e. not most abundant in all sites), 0-1

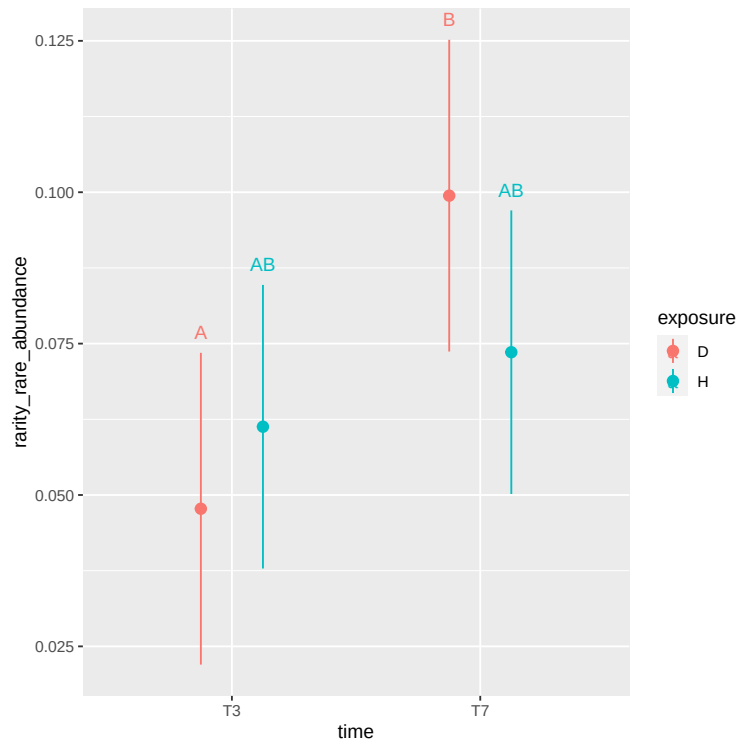
#for healthy samples, rare species increased in abundance at T7

anova(r_alpha_models$model[[7]])

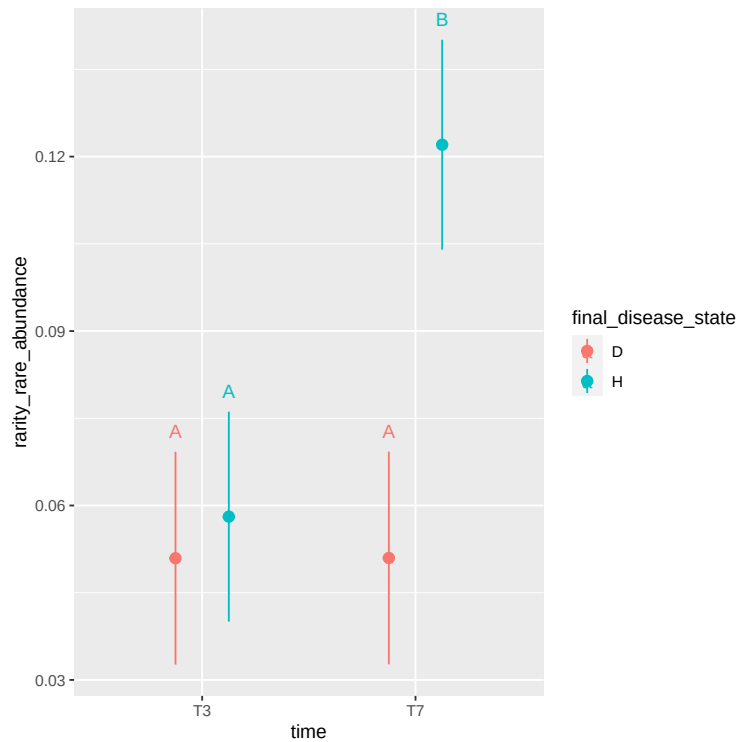
## Type III Analysis of Variance Table with Satterthwaite's method
##
##          Sum Sq  Mean Sq NumDF  DenDF F value    Pr(>F)
## exposure      0.000039 0.000039     1    3.311   0.0322   0.86803
## final_disease_state 0.024400 0.024400     1   89.646  20.1021 2.165e-05 ***
## time          0.045466 0.045466     1  93.000  37.4580 2.197e-08 ***
## exposure:time    0.008363 0.008363     1  93.000   6.8901  0.01013 *
## final_disease_state:time 0.021241 0.021241     1  93.000  17.4997 6.505e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#final_disease_state ***
#time ***

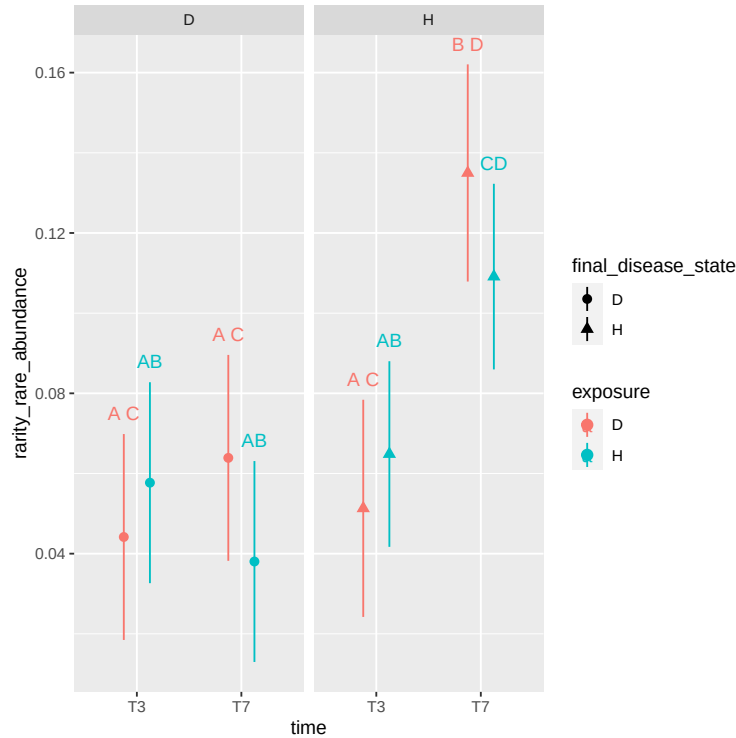
#exposure:time *
emmeans(r_alpha_models$model[[7]], ~(exposure) * time,
        type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
             col = exposure)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[7])
```



```
#final_disease_state:time ***
emmeans(r_alpha_models$model[[7]], ~(final_disease_state) * time,
         type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
             col = final_disease_state)) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[7])
```



```
#all
emmeans(r_alpha_models$model[[7]], ~(exposure + final_disease_state) * time,
         type = 'response') %>%
  cld(Letters = LETTERS) %>%
  as_tibble() %>%
  mutate(.group = str_trim(.group)) %>%
  #rename(emmean = response) %>%
  ggplot(aes(x = time, y = emmean, ymin = emmean - SE, ymax = emmean + SE,
             col = exposure, shape = final_disease_state)) +
  facet_wrap(~final_disease_state) +
  geom_pointrange(position = position_dodge(0.5)) +
  geom_text(aes(y = (emmean + SE), label = .group),
            position = position_dodge(0.5), vjust = -1) +
  ylab(r_alpha_models$metric[7])
```



#buffer line

Alpha Diversity Index Notes

#richness

#observed is observed species richness

#chao1 is nonparametric method for estimating the number of species in a community,

#based on the concept that rare species infer the most information about the number of missing species

#particularly useful for data sets skewed toward the low-abundance species

#dominance

#dbp is the relative abundance of the most abundant species of the sample.

#Index gives values in interval 0 to 1, where bigger value represent greater dominance

#dmn is the sum of relative abundances of the two most abundant species of the sample

#absolute index equals to the absolute abundance of the most dominant n species of the sample

#(specify the number with the argument ntaxa)

#relative index equals to the relative abundance of the most dominant n species of the sample

#(specify the number with the argument ntaxa). This index gives values in interval 0 to 1

#simpson's lambda is the probability that two randomly chosen individuals belongs to the same species.

#The higher the probability, the greater the dominance

#core_abundance is the sum of relative abundances of core species in the sample.

#Index gives values in interval 0 to 1, where bigger value represent greater dominance

#Core species are species that are most abundant in all samples

#gini measures how unevenly abundances are distributed

#If there is small group of species that represent large portion of total abundance of microbes,

#the inequality is large and Gini index closer to 1. If all species has equally large abundances,

#the equality is perfect and Gini index equals 0

#Rarity and low abundance

*#log_modulo_skewness is a rarity index that characterizes the concentration of species at low abundance
#It uses the skewness of the frequency distribution of arithmetic abundance classes
#low abundance gives the concentration of species at low abundance, or the relative proportion of rare
#species in [0,1].The species that are below the indicated detection threshold are considered rare
#use "detection = " (ex: detection = 0.2/100)
#Note that population prevalence is not considered. If the detection argument is a vector,
#then a data.frame is returned, one column for each detection threshold.
#rare abundance gives the relative proportion of rare species in the interval [0,1].
#(rare = those that are not part of the core microbiota)
#This is the complement (1-x) of the core abundance. The rarity function provides the
#abundance of the least abundant taxa within each sample, regardless of the population prevalence.*

#Diversity

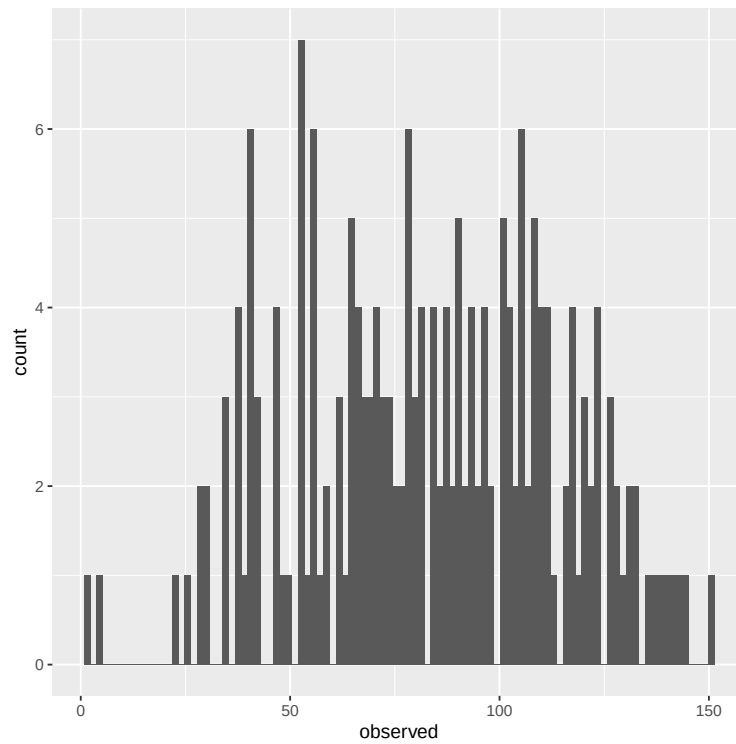
*#inverse_simpson is an indication of the richness in a community with uniform evenness that would have
#the same level of diversity, calculated as 1/lambda where lambda is the simpson index
#gini-simpson measures the probability that two randomly selected individuals belong to different species
#1 - lambda where lambda is the simpson index
#shannon shows how diverse the species in a given community are
#It rises with the number of species and the evenness of their abundance
#fisher's alpha describes mathematically the relationship between the number of species
#and the number of individuals in those species
#coverage gives the number of groups needed to have a given proportion of the ecosystem occupied
#(by default is 0.5 ie 50%)*

#Evenness

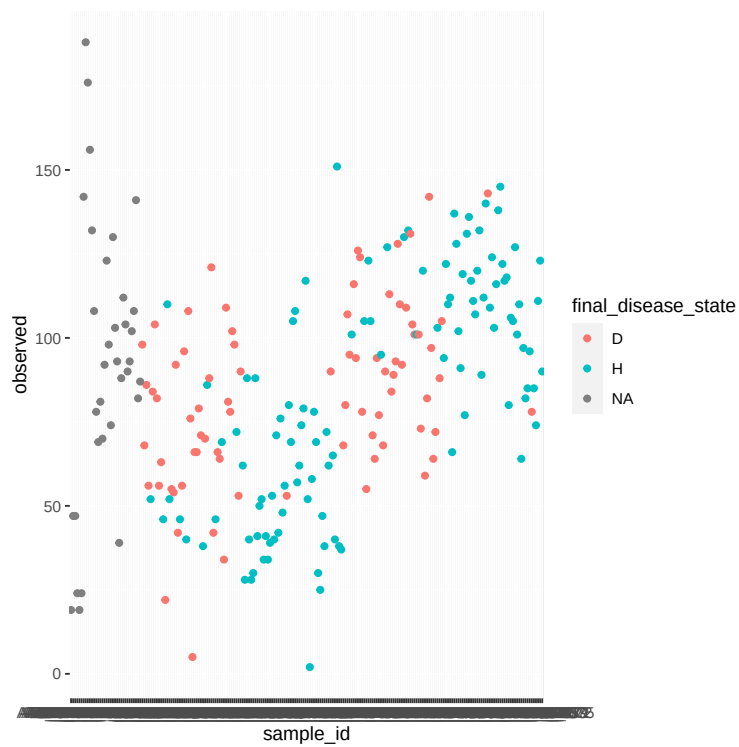
*#camargo's compares proportions of individuals between sites, with 1 being even and 0 being patchy
#relatively unaffected by sites with very few organisms, and is unaffected by site richness
#simpson's evenness is a variant of the reciprocal Simpson index
#Index values range from near 0 (1/s) (patchy or skewed) to 1 (even), and the index
#is relatively unaffected by sites with very few individuals.
#pielou is shannon diversity index value divided by the maximum possible shannon diversity index given
#complete evenness (proportion 0 to 1)
#evvar is based on the variance in abundance over the species taken over log abundance
#so proportional differences are compared, then converted to a 0-1 scale by arctan
#bulla gives equal weight to all species regardless of abundance so it's
#sensitive to the presence of rare species*

exploratory plots

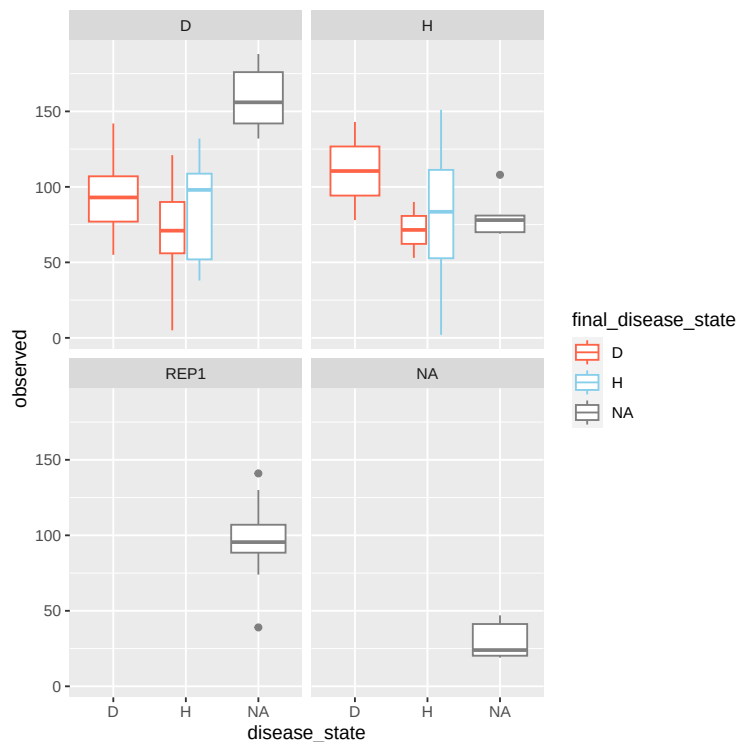
```
timepoint_data %>%  
  ggplot(aes(x = observed)) +  
  geom_histogram(bins = 100)
```



```
alpha_table %>%
  arrange(exposure) %>%
  ggplot() +
    geom_point(aes(x = sample_id, y = observed, col = final_disease_state))
```

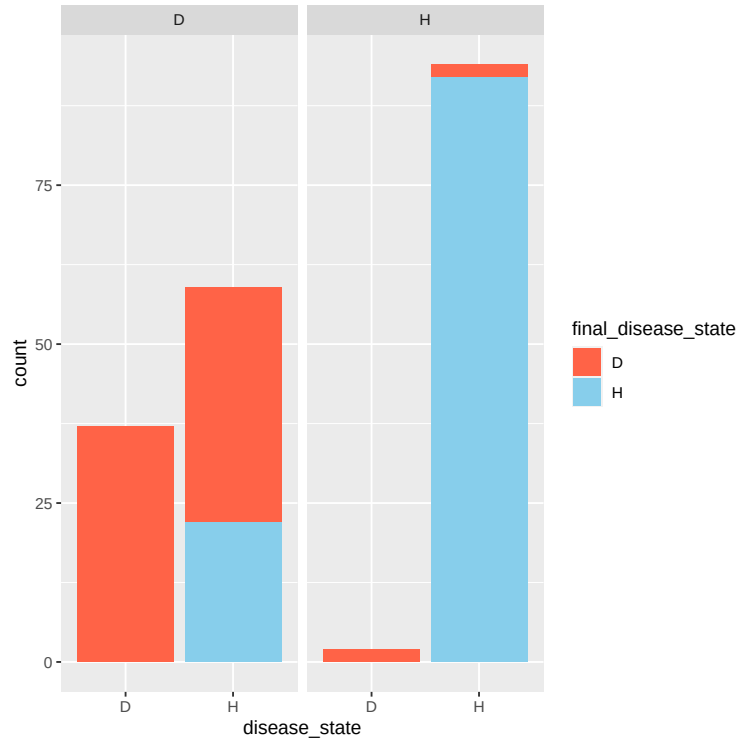


```
ggplot(data = alpha_table) +
  geom_boxplot(aes(x = disease_state, y = observed, col = final_disease_state)) +
  scale_color_manual(values = c("tomato", "skyblue")) +
  facet_wrap(~exposure)
```

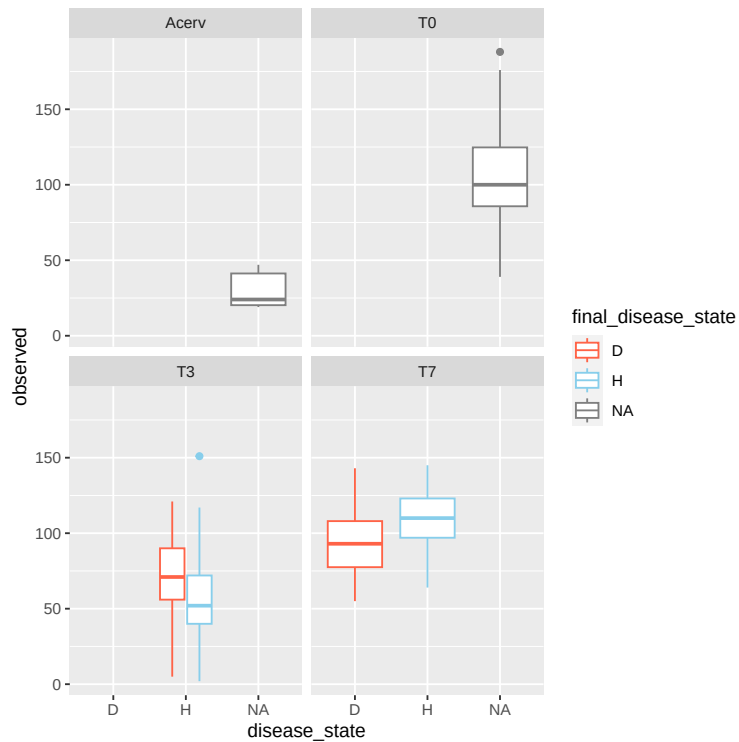


```
ggplot(data = timepoint_data) +
  geom_histogram(aes(x = disease_state, fill = final_disease_state), stat="count") +
  scale_fill_manual(values = c("tomato", "skyblue")) +
  facet_wrap(~exposure)
```

```
## Warning in geom_histogram(aes(x = disease_state, fill = final_disease_state), : Ignoring
unknown parameters: 'binwidth', 'bins', and 'pad'
```

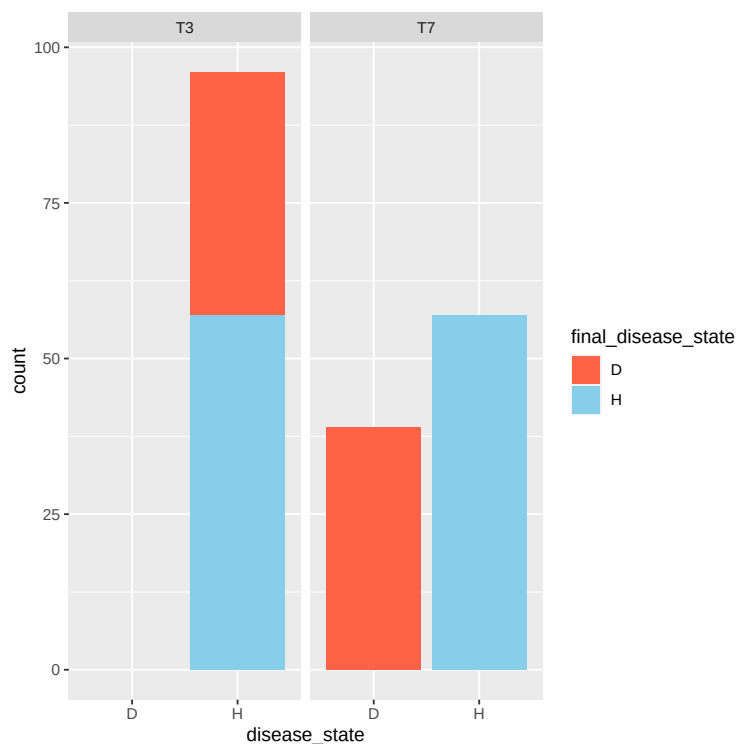


```
ggplot(data = alpha_table) +
  geom_boxplot(aes(x = disease_state, y = observed, col = final_disease_state)) +
  scale_color_manual(values = c("tomato", "skyblue")) +
  facet_wrap(~time)
```

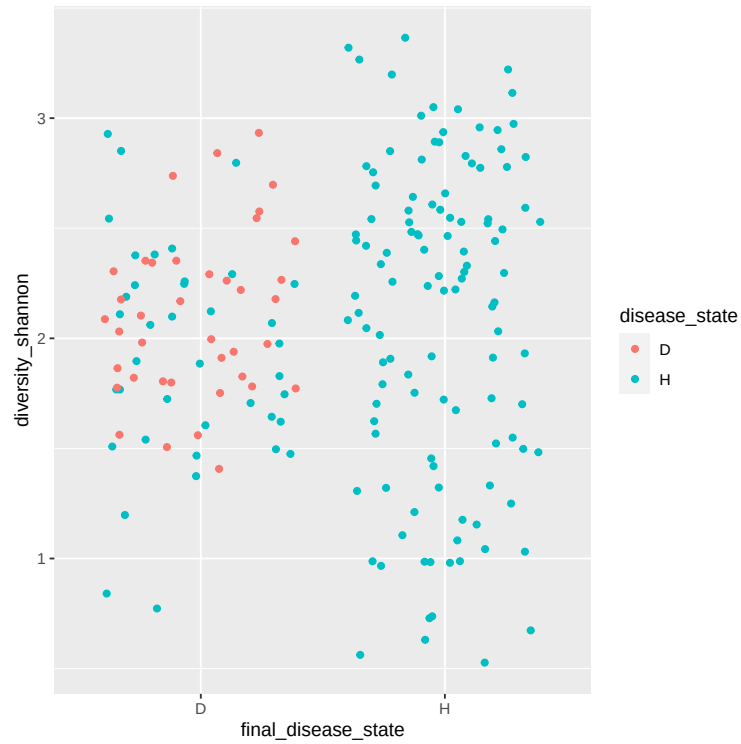


```
ggplot(data = timepoint_data) +
  geom_histogram(aes(x = disease_state, fill = final_disease_state), stat="count") +
  scale_fill_manual(values = c("tomato", "skyblue")) +
  facet_wrap(~time)

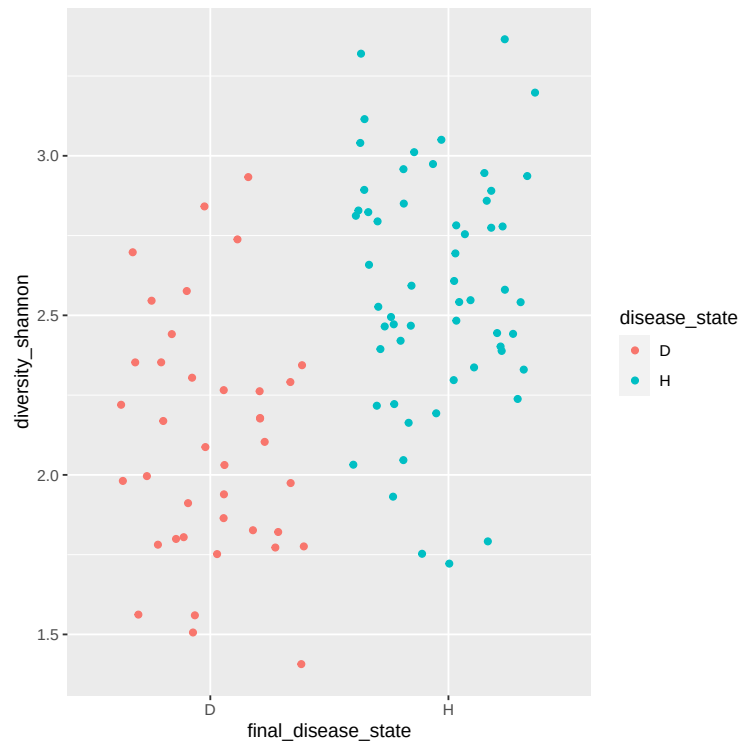
## Warning in geom_histogram(aes(x = disease_state, fill = final_disease_state), : Ignoring
unknown parameters: 'binwidth', 'bins', and 'pad'
```



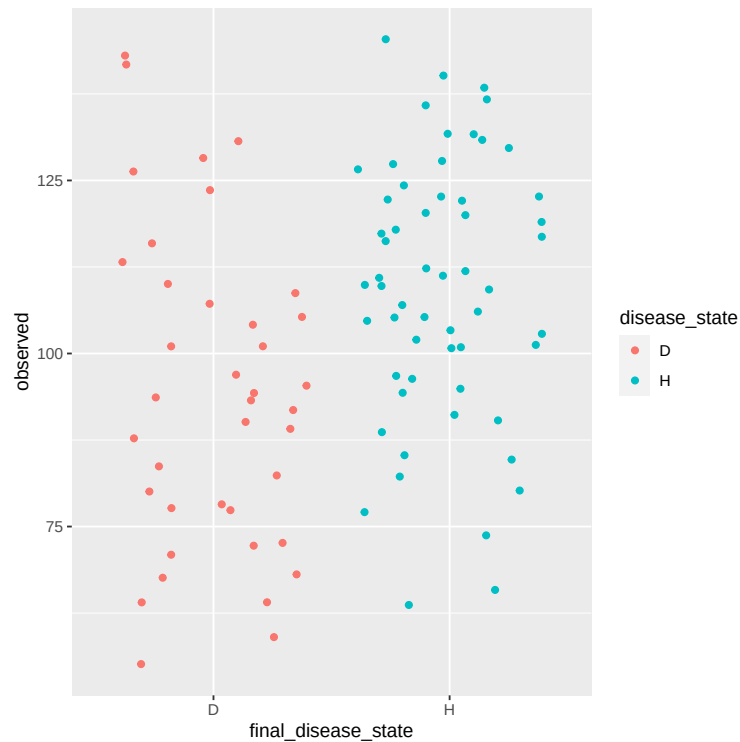
```
ggplot(data = timepoint_data) +
  geom_jitter(aes(x = final_disease_state, y = diversity_shannon, col = disease_state))
```



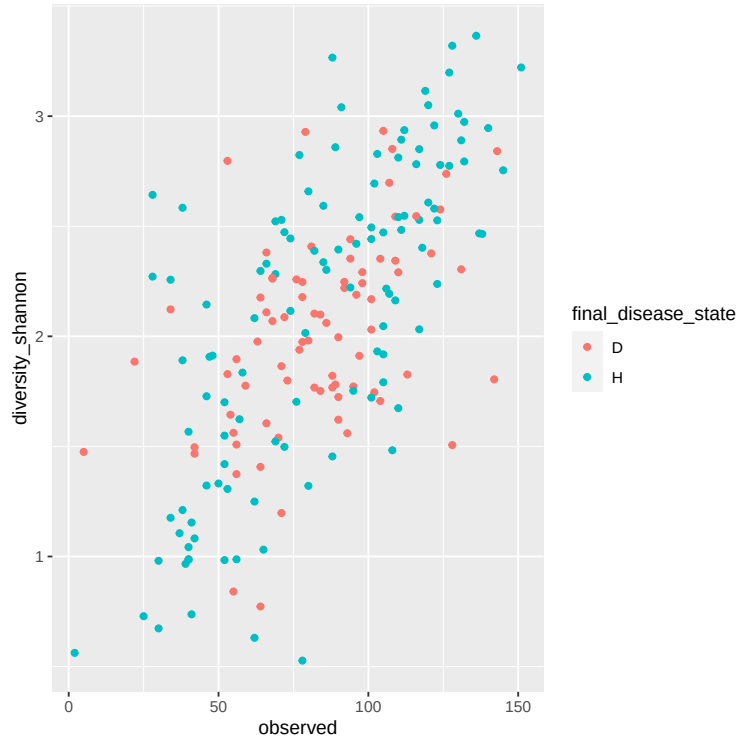
```
timepoint_data %>%
  filter(time == "T7") %>%
  ggplot() +
    geom_jitter(aes(x = final_disease_state, y = diversity_shannon, col = disease_state))
```



```
timepoint_data %>%
  filter(time == "T7") %>%
  ggplot() +
  geom_jitter(aes(x = final_disease_state, y = observed, col = disease_state))
```



```
ggplot(data = timepoint_data) +
  geom_point(aes(x = observed, y = diversity_shannon, col = final_disease_state)) #+
```



```
geom_smooth(method = "lm", se = TRUE)

## geom_smooth: na.rm = FALSE, orientation = NA, se = TRUE
## stat_smooth: na.rm = FALSE, orientation = NA, se = TRUE, method = lm
## position_identity
#### ####
```

The R session information (including the OS info, R version and all packages used):

```
sessionInfo()

## R version 4.2.1 (2022-06-23)
## Platform: x86_64-apple-darwin17.0 (64-bit)
## Running under: macOS Monterey 12.5
##
## Matrix products: default
## LAPACK: /Library/Frameworks/R.framework/Versions/4.2/Resources/lib/libRlapack.dylib
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] forcats_1.0.0    stringr_1.5.0    dplyr_1.1.0      purrr_1.0.1
## [5] readr_2.1.3      tidyr_1.3.0      tibble_3.1.8     tidyverse_1.3.2
## [9] multcomp_1.4-20  TH.data_1.1-1    MASS_7.3-58.2    survival_3.5-0
## [13] mvtnorm_1.1-3    car_3.1-1        carData_3.0-5    emmeans_1.8.4-1
```

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## [17] afex_1.2-1      lme4_1.1-31      Matrix_1.5-3      vegan_2.6-4
## [21] lattice_0.20-45 permute_0.9-7     microbiome_1.20.0 ggplot2_3.4.0
## [25] phyloseq_1.42.0
##
## loaded via a namespace (and not attached):
## [1] googledrive_2.0.0      Rtsne_0.16        minqa_1.2.5
## [4] colorspace_2.1-0       ellipsis_0.3.2     estimability_1.4.1
## [7] snakecase_0.11.0       XVector_0.38.0     fs_1.6.1
## [10] rstudioapi_0.14        farver_2.1.1       fansi_1.0.4
## [13] lubridate_1.9.1        xml2_1.3.3         codetools_0.2-19
## [16] splines_4.2.1          knitr_1.42         ade4_1.7-20
## [19] jsonlite_1.8.4         nloptr_2.0.3       pbkrtest_0.5.2
## [22] broom_1.0.3            cluster_2.1.4      dbplyr_2.3.0
## [25] httr_1.4.4            compiler_4.2.1     backports_1.4.1
## [28] assertthat_0.2.1       gargle_1.3.0       limma_3.54.1
## [31] cli_3.6.0             tools_4.2.1        lmerTest_3.1-3
## [34] igraph_1.3.5          gtable_0.3.1       glue_1.6.2
## [37] GenomeInfoDbData_1.2.9 reshape2_1.4.4     Rcpp_1.0.10
## [40] Biobase_2.58.0         cellranger_1.1.0   vctrs_0.5.2
## [43] Biostrings_2.66.0      rhdf5filters_1.10.0 multtest_2.54.0
## [46] ape_5.6-2             nlme_3.1-162       iterators_1.0.14
## [49] xfun_0.37             rvest_1.0.3        timechange_0.2.0
## [52] lifecycle_1.0.3        googlesheets4_1.0.1 edgeR_3.40.2
## [55] zlibbioc_1.44.0        zoo_1.8-11         scales_1.2.1
## [58] hms_1.1.2            parallel_4.2.1     biomformat_1.26.0
## [61] sandwich_3.0-2        rhdf5_2.42.0       stringi_1.7.12
## [64] highr_0.10           S4Vectors_0.36.1   foreach_1.5.2
## [67] BiocGenerics_0.44.0    boot_1.3-28.1      GenomeInfoDb_1.34.9
## [70] rlang_1.0.6           pkgconfig_2.0.3     bitops_1.0-7
## [73] evaluate_0.20         Rhdf5lib_1.20.0     labeling_0.4.2
## [76] tidyselect_1.2.0      plyr_1.8.8         magrittr_2.0.3
## [79] R6_2.5.1             multcompView_0.1-8  IRanges_2.32.0
## [82] generics_0.1.3        DBI_1.1.3          haven_2.5.1
## [85] pillar_1.8.1         withr_2.5.0        mgcv_1.8-41
## [88] abind_1.4-5          RCurl_1.98-1.10     modelr_0.1.10
## [91] janitor_2.2.0         crayon_1.5.2       utf8_1.2.3
## [94] tzdb_0.3.0           readxl_1.4.1       locfit_1.5-9.7
## [97] grid_4.2.1           data.table_1.14.6   reprex_2.0.2
## [100] xtable_1.8-4         numDeriv_2016.8-1.1 stats4_4.2.1
## [103] munsell_0.5.0
##
## Sys.time()
## [1] "2023-02-24 10:51:28 EST"

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